Lab 3

Screenshots for outputs and commands run

```
root@d0ade0c713f7: /
9:perf_event:/docker/d0ade0c713f73c1afb51bf09f9de73c7245c9bd0d9946edc23864f6a967680a7
8:net_cls:/docker/d0ade0c713f73c1afb51bf09f9de73c7245c9bd0d9946edc23864f6a967680a7
7:freezer:/docker/d0ade0c713f73c1afb51bf09f9de73c7245c9bd0d9946edc23864f6a967680a7
6:devices:/docker/d0ade0c713f73c1afb51bf09f9de73c7245c9bd0d9946edc23864f6a967680a7
5:memory:/docker/d0ade0c713f73c1afb51bf09f9de73c7245c9bd0d9946edc23864f6a967680a7
4:blkio:/docker/d0ade0c713f73c1afb51bf09f9de73c7245c9bd0d9946edc23864f6a967680a7
3:cpuacct:/docker/d0ade0c713f73c1afb51bf09f9de73c7245c9bd0d9946edc23864f6a967680a7
2:cpu:/docker/d0ade0c713f73c1afb51bf09f9de73c7245c9bd0d9946edc23864f6a967680a7
1:cpuset:/docker/d0ade0c713f73c1afb51bf09f9de73c7245c9bd0d9946edc23864f6a967680a7
0::/docker/d0ade0c713f73c1afb51bf09f9de73c7245c9bd0d9946edc23864f6a967680a7
root@d0ade0c713f7:/# hostname
d0ade0c713f7
root@d0ade0c713f7:/# ps -ef
UID        PID  PPID  C STIME TTY          TIME CMD
root         1     0  0 05:13 pts/0    00:00:00 bash
root        16     1  0 05:14 pts/0    00:00:00 ps -ef
root@d0ade0c713f7:/# ip addr
bash: ip: command not found
root@d0ade0c713f7:/# mount | head
overlay on / type overlay (rw,relatime,lowerdir=/var/lib/desktop-containerd/daemon/io.containerd.snaps
hotter.v1.overlayfs/snapshots/566/fs:/var/lib/desktop-containerd/daemon/io.containerd.snapshotter.v1.o
verlayfs/snapshots/565/fs,upperdir=/var/lib/desktop-containerd/daemon/io.containerd.snapshotter.v1.ove
rlayfs/snapshots/567/fs,workdir=/var/lib/desktop-containerd/daemon/io.containerd.snapshotter.v1.overla
yfs/snapshots/567/work)
proc on /proc type proc (rw,nosuid,nodev,noexec,relatime)
tmpfs on /dev type tmpfs (rw,nosuid,size=65536k,mode=755)
devpts on /dev/pts type devpts (rw,nosuid,noexec,relatime,gid=5,mode=620,ptmxmode=666)
sysfs on /sys type sysfs (ro,nosuid,nodev,noexec,relatime)
tmpfs on /sys/fs/cgroup type tmpfs (rw,nosuid,nodev,noexec,relatime,mode=755)
cpuset on /sys/fs/cgroup/cpuset type cgroup (ro,nosuid,nodev,noexec,relatime,cpuset)
cpu on /sys/fs/cgroup/cpu type cgroup (ro,nosuid,nodev,noexec,relatime,cpu)
cpuacct on /sys/fs/cgroup/cpuacct type cgroup (ro,nosuid,nodev,noexec,relatime,cpuacct)
blkio on /sys/fs/cgroup/blkio type cgroup (ro,nosuid,nodev,noexec,relatime,blkio)
root@d0ade0c713f7:/# cat /proc/1/cgroup
15:name=systemd:/docker/d0ade0c713f73c1afb51bf09f9de73c7245c9bd0d9946edc23864f6a967680a7
14:misc:/docker/d0ade0c713f73c1afb51bf09f9de73c7245c9bd0d9946edc23864f6a967680a7
13:rdma:/docker/d0ade0c713f73c1afb51bf09f9de73c7245c9bd0d9946edc23864f6a967680a7
12:pids:/docker/d0ade0c713f73c1afb51bf09f9de73c7245c9bd0d9946edc23864f6a967680a7
11:hugetlb:/docker/d0ade0c713f73c1afb51bf09f9de73c7245c9bd0d9946edc23864f6a967680a7
10:net_prio:/docker/d0ade0c713f73c1afb51bf09f9de73c7245c9bd0d9946edc23864f6a967680a7
9:perf_event:/docker/d0ade0c713f73c1afb51bf09f9de73c7245c9bd0d9946edc23864f6a967680a7
8:net_cls:/docker/d0ade0c713f73c1afb51bf09f9de73c7245c9bd0d9946edc23864f6a967680a7
7:freezer:/docker/d0ade0c713f73c1afb51bf09f9de73c7245c9bd0d9946edc23864f6a967680a7
6:devices:/docker/d0ade0c713f73c1afb51bf09f9de73c7245c9bd0d9946edc23864f6a967680a7
5:memory:/docker/d0ade0c713f73c1afb51bf09f9de73c7245c9bd0d9946edc23864f6a967680a7
4:blkio:/docker/d0ade0c713f73c1afb51bf09f9de73c7245c9bd0d9946edc23864f6a967680a7
3:cpuacct:/docker/d0ade0c713f73c1afb51bf09f9de73c7245c9bd0d9946edc23864f6a967680a7
2:cpu:/docker/d0ade0c713f73c1afb51bf09f9de73c7245c9bd0d9946edc23864f6a967680a7
1:cpuset:/docker/d0ade0c713f73c1afb51bf09f9de73c7245c9bd0d9946edc23864f6a967680a7
0::/docker/d0ade0c713f73c1afb51bf09f9de73c7245c9bd0d9946edc23864f6a967680a7
root@d0ade0c713f7:/#
```

```
C:\WINDOWS\system32\cmd.

C:\Users\DELL>docker stats
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT   MEM %     NET I/O           BLOCK I/O   PIDS
139e148e3633   sad_swirles   0.12%     18.96MiB / 256MiB   7.41%     47.6MB / 812kB    0B / 0B     1
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT   MEM %     NET I/O           BLOCK I/O   PIDS
139e148e3633   sad_swirles   0.00%     18.96MiB / 256MiB   7.41%     47.6MB / 812kB    0B / 0B     1
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```
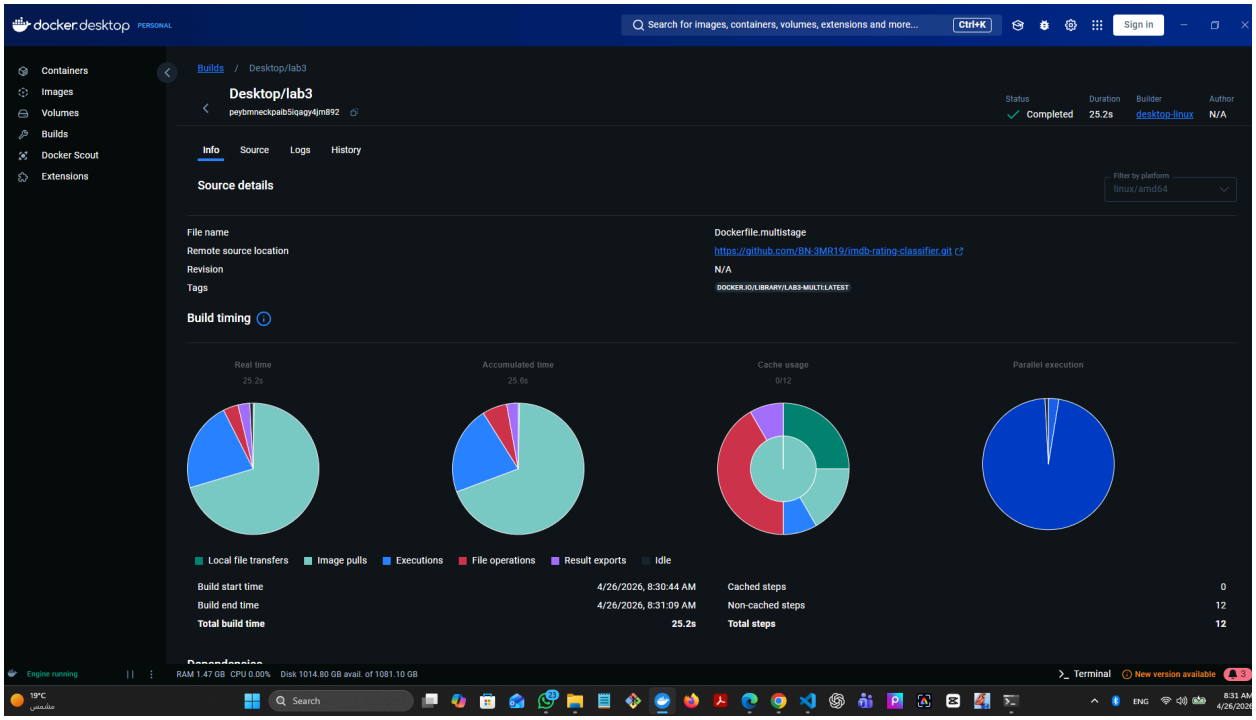
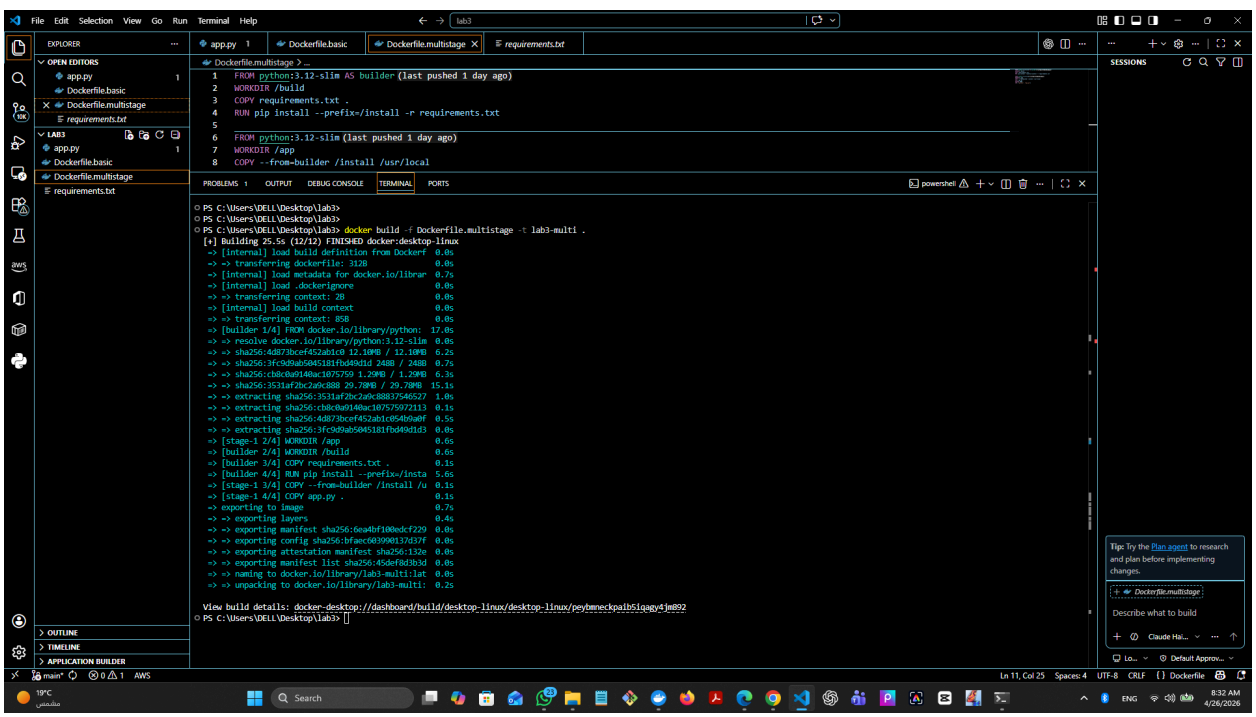

```
File  Edit  Selection  View  Go  Run  Terminal  Help        lab3

EXPLORER
OPEN EDITORS
  Welcome
  app.py 1
  Dockerfile.basic
LAB3
  app.py 1
  Dockerfile.basic

Welcome | app.py 1 | Dockerfile.basic
Dockerfile.basic > ...
1  FROM python:3.12-slim (last pushed 1 day ago)
2  WORKDIR /app
3  COPY requirements.txt .
4  RUN pip install -r requirements.txt
5  COPY app.py .
6  EXPOSE 5000
7  CMD ["python", "app.py"]

PROBLEMS 1   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS        powershell

  View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/7252w34098gu41q26ncwd1821b
  PS C:\Users\DELL\Desktop\lab3> docker image ls
>> docker history lab3-basic
  REPOSITORY                      TAG       IMAGE ID       CREATED        SIZE
  redis                           7         3e0669e42d4f   4 days ago     173MB
  ubuntu                          22.04     962f6cadeae0   2 weeks ago    117MB
  ubuntu                          latest    c4a8d5503dfb   2 weeks ago    117MB
  adventureworksdocker-api        latest    af7a7b1555e4   16 months ago  363MB
  webapplication1-api             latest    251aedf8eb3d   16 months ago  351MB
  restapi                         dev       eb4d4d1fd67f   16 months ago  328MB
  grpcservice                     dev       f39e055eebb8   16 months ago  328MB
  webapplication1                 dev       db9348e6a2c5   16 months ago  328MB
  adventureworksapi-api           latest    07b14fd4b18c   16 months ago  370MB
  mcr.microsoft.com/dotnet/aspnet 9.0       d8f01f752bf9   16 months ago  328MB
  proj999-api                     latest    4b5e93d9a37c   17 months ago  310MB
  productapi-api                  latest    e89f298d6490   17 months ago  313MB
  jenkins/jenkins                 latest    4bc5685ff6e6   17 months ago  779MB
  archproj1-server2               latest    b4fb3421faf0   17 months ago  208MB
  archproj1-server1               latest    0b22fab134b8   17 months ago  208MB
  project-server1                 latest    b6efa7746c4a   17 months ago  208MB
  project-server2                 latest    a9a9608d67d9   17 months ago  208MB
  grafana/grafana                 latest    fa081ab6e1ae   17 months ago  660MB
  prom/prometheus                 latest    3b9b2a15d376   17 months ago  410MB
  mcr.microsoft.com/mssql/server  latest    45a1a9d13ca5   17 months ago  2.19GB
  fedora                          latest    3ec60eb34fa1   18 months ago  235MB
  redis                           latest    af0be38eb8e4   18 months ago  174MB
  redis                           alpine    c1d8045c8b52   18 months ago  68.2MB
  quay.io/coreos/etcd             v3.5.0    28759af54acd   4 years ago    156MB
  Error response from daemon: No such image: lab3-basic:latest
PS C:\Users\DELL\Desktop\lab3>

OUTLINE
TIMELINE
APPLICATION BUILDER

SESSIONS
Tip: Try the Plan agent to research and plan before implementing changes.
Dockerfile.basic
Describe what to build
Claude Hai...
Lo...   Default Approv...

main*   0 1   AWS   Ln 7, Col 25   Spaces: 4   UTF-8   CRLF   Dockerfile
```


docker desktop PERSONAL
Search for images, containers, volumes, extensions and more...
Ctrl+K
Sign in
Containers
Images
Volumes
Builds
Docker Scout
Extensions
Builds / Desktop/lab3
Desktop/lab3
peybmneckpaib5iqagy4jm892
Status
Completed
Duration
25.2s
Builder
desktop-linux
Author
N/A
Info
Source
Logs
History
Source details
Filter by platform
linux/amd64
File name
Dockerfile.multistage
Remote source location
https://github.com/BN-3MR19/imdb-rating-classifier.git
Revision
N/A
Tags
DOCKER.IO/LIBRARY/LAB3-MULTI:LATEST
Build timing
Real time
25.2s
Accumulated time
25.6s
Cache usage
0/12
Parallel execution
Local file transfers
Image pulls
Executions
File operations
Result exports
Idle
Build start time
4/26/2026, 8:30:44 AM
Build end time
4/26/2026, 8:31:09 AM
Total build time
25.2s
Cached steps
0
Non-cached steps
12
Total steps
12
Engine running
RAM 1.47 GB CPU 0.00% Disk 1014.80 GB avail. of 1081.10 GB
Terminal
New version available


File Edit Selection View Go Run Terminal Help
lab3
EXPLORER
OPEN EDITORS
app.py
Dockerfile.basic
Dockerfile.multistage
requirements.txt
LAB3
app.py
Dockerfile.basic
Dockerfile.multistage
requirements.txt
OUTLINE
TIMELINE
APPLICATION BUILDER
Dockerfile.multistage > ...
1 FROM python:3.12-slim AS builder (last pushed 1 day ago)
2 WORKDIR /build
3 COPY requirements.txt .
4 RUN pip install --prefix=/install -r requirements.txt
5
6 FROM python:3.12-slim (last pushed 1 day ago)
7 WORKDIR /app
8 COPY --from=builder /install /usr/local
PROBLEMS 1
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
powershell
PS C:\Users\DELL\Desktop\lab3>
PS C:\Users\DELL\Desktop\lab3>
PS C:\Users\DELL\Desktop\lab3> docker build -f Dockerfile.multistage -t lab3-multi .
[+] Building 25.5s (12/12) FINISHED docker:desktop-linux
=> [internal] load build definition from Dockerf 0.0s
=> => transferring dockerfile: 312B 0.0s
=> [internal] load metadata for docker.io/librar 0.7s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [internal] load build context 0.0s
=> => transferring context: 85B 0.0s
=> [builder 1/4] FROM docker.io/library/python: 17.0s
=> => resolve docker.io/library/python:3.12-slim 0.0s
=> => sha256:4d873bcef452ab1c0 12.10MB / 12.10MB 6.2s
=> => sha256:3fc9d9ab5045181fbd49d1d 248B / 248B 0.7s
=> => sha256:cb0c0a9140ac1075759 1.29MB / 1.29MB 6.3s
=> => sha256:3531af2bc2a9c888 29.78MB / 29.78MB 15.1s
=> => extracting sha256:3531af2bc2a9c88837546527 1.0s
=> => extracting sha256:cb0c0a9140ac107575972113 0.1s
=> => extracting sha256:4d873bcef452ab1c0540b6a0f 0.5s
=> => extracting sha256:3fc9d9ab5045181fbd49d1d3 0.0s
=> [stage-1 2/4] WORKDIR /app 0.6s
=> [builder 2/4] WORKDIR /build 0.6s
=> [builder 3/4] COPY requirements.txt . 0.1s
=> [builder 4/4] RUN pip install --prefix=/insta 5.6s
=> [stage-1 3/4] COPY --from=builder /install /u 0.1s
=> [stage-1 4/4] COPY app.py . 0.1s
=> exporting to image 0.7s
=> => exporting layers 0.4s
=> => exporting manifest sha256:6ea4bf100edcf229 0.0s
=> => exporting config sha256:bfaec60399013 7d37f 0.0s
=> => exporting attestation manifest sha256:132e 0.0s
=> => exporting manifest list sha256:45de f8d3b3d 0.0s
=> => naming to docker.io/library/lab3-multi:lat 0.0s
=> => unpacking to docker.io/library/lab3-multi: 0.2s
View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/peybmneckpaib5iqagy4jm892
PS C:\Users\DELL\Desktop\lab3>
SESSIONS
Tip: Try the Plan agent to research and plan before implementing changes.
Dockerfile.multistage
Describe what to build
Claude Hai...
Default Approv...
main*
AWS
Ln 11, Col 25
Spaces: 4
UTF-8
CRLF
Dockerfile

SHORT & PERFECT answers

Namespaces alone only isolate processes but don't limit resource usage → processes can still consume unlimited CPU/memory.

Cgroups enforce limits on CPU, memory, etc. → preventing one container from starving others → improves stability.

Image layering enables caching, reuse, and smaller updates → faster builds, less storage, efficient deployment at scale.

Desired state means Kubernetes maintains the system to match a defined configuration automatically (e.g., always 3 running Pods).

Self-healing is automatic recovery (restart/replace Pods) → unlike manual intervention in traditional systems.

Readiness probe = is app ready to serve traffic

Liveness probe = is app still alive

→ different purposes, not interchangeable.

Single-node limitation: no real scheduling decisions → cannot observe distribution, load balancing, or multi-node placement behavior.